



Monogenic Diabetes (MODY & Neonatal Diabetes Mellitus)

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What is monogenic diabetes?

Monogenic diabetes is a group of uncommon forms of **diabetes** that are caused by a **variant** , or change, in a single **gene**. Researchers have found variants in more than 20 genes that cause monogenic diabetes.¹ Monogenic forms of diabetes are part of the group of atypical diabetes. People with atypical forms of diabetes have symptoms and signs that are different than those of **type 1** or **type 2** diabetes.

If you have monogenic diabetes, you may have high levels of blood glucose, also called blood sugar. Diabetes may raise your risk of eye, [kidney](#), nerve, and heart damage. Diabetes is also linked to some types of cancer. You may be able to prevent or delay [diabetes health problems](#) by [managing your blood glucose level](#).

Maturity-onset diabetes of the young (MODY) and neonatal diabetes mellitus (NDM) are the two main forms of monogenic diabetes.

What is maturity-onset diabetes of the young (MODY)?

MODY forms of diabetes usually develop in teens or young adults. People with some forms of MODY have blood glucose levels that are only a little higher than normal and do not change much during their lives. The levels of blood glucose are typically not high enough to damage their organs, so they may not need treatment. However, people with other forms of MODY have higher blood glucose levels and need treatment.

What is neonatal diabetes mellitus (NDM)?

NDM forms of diabetes develop in the first 6 to 12 months of life.^{2,3} Babies with forms of NDM do not produce enough [insulin](#), which allows their blood glucose levels to go up more than normal. Babies who develop diabetes in the first 6 months of life almost always have a form of NDM.²

In most forms of NDM, babies will have lifelong diabetes. These forms of NDM are called permanent NDM. Babies with some forms of NDM have diabetes as infants, but their diabetes goes away during childhood. For these babies, diabetes often shows up again later in life. These forms of NDM are called transient NDM.

How common is monogenic diabetes?

Some research, mainly from the United States and Europe, has found that

- about 1 to 5 of every 100 people with diabetes have monogenic diabetes^{4,5}
- NDM is a rare disease, and only 1 in every 90,000 babies has the disease⁴

People with monogenic diabetes may be misdiagnosed with [type 1](#) or [type 2](#) diabetes because the symptoms may be similar. Therefore, it is difficult to know how many people have monogenic diabetes.

Who is more likely to have monogenic diabetes?

People with MODY forms of diabetes usually have other family members with diabetes. MODY forms of diabetes most often occur in people ages 30 or younger.⁶ Forms of NDM most often occur in babies ages 6 months or younger.^{1,2} More studies are needed to understand which populations, based on race, are more affected by forms of MODY and NDM.

What health problems can people with monogenic diabetes develop?

Some people with forms of monogenic diabetes may develop the same diabetes **health problems** as people with other types of diabetes. High blood glucose levels from diabetes can damage every part of your body, including your

- heart
- kidneys
- nerves
- eyes
- feet
- teeth

If you have a form of monogenic diabetes, work with your primary health care professional, who may be a doctor, physician assistant, or nurse practitioner, to manage your diabetes. Your health care team can help you lower your chances of developing other health problems.

What are the symptoms of monogenic diabetes?

The symptoms of monogenic diabetes depend on which gene variant you have. Some people have no symptoms. Other people may have general symptoms of diabetes that include

- increased urination
- feeling thirsty
- feeling very hungry, even after you have eaten
- blurred vision

- fatigue
- numbness or tingling in the feet or hands
- sores that do not heal

What causes monogenic diabetes?

A variant, or change, in a single gene causes monogenic diabetes.

You inherited genes—which determine things like your hair color, eye color, and height—from each of your parents. In most cases of MODY forms of diabetes, you inherit a gene from one or both parents that causes the disease.

Babies with NDM forms of diabetes rarely have a family history of the disease. Most cases of NDM are caused by a gene change that happens while a baby develops in the womb.

Most people with MODY forms of diabetes have inherited a gene for the disease from a parent who also had the disease. Diseases caused by a gene from one parent are called [autosomal dominant](#) [NIH](#) . If you have a form of NDM or MODY that is autosomal dominant, your child has a 50% chance of having the disease.

Some forms of NDM or MODY occur only if a child inherits a gene for the disease from both parents. In most cases, neither of the child's parents had the disease. Diseases that occur only if you inherit a gene from both parents who don't have the disease are called [autosomal recessive](#) [NIH](#) . If you inherit a gene for a recessive form of NDM or MODY from only one parent, you won't have the disease. However, you can pass that gene to future generations.

Different gene variants cause monogenic diabetes in different ways. For example, gene variants may

- affect how the pancreas develops
- cause the pancreas to produce insulin that does not work well
- destroy cells in the pancreas that produce insulin

How do health care professionals diagnose monogenic diabetes?

Your health care professional will usually test you for forms of monogenic diabetes if you have diabetes, are younger than 30 years old, don't have [overweight or obesity](#), and have family members with these forms of diabetes.

Health care professionals will diagnose diabetes using your symptoms, medical and family history, and [diabetes tests](#). Your health care professional may also order tests to find out if your diabetes is type 1, type 2, or monogenic. The type of diabetes you have will help your health care professional determine how to treat the disease.

Blood glucose tests

Health care professionals can diagnose diabetes using blood tests that measure your glucose level to see if it is higher than normal. People with some forms of monogenic diabetes have blood glucose levels that are a little higher than normal.²

Blood glucose tests can help your health care professional tell if you have diabetes. However, these tests do not show the type of diabetes you have.

Autoantibodies

Your health care professional will order a test for autoantibodies to find out if you have type 1 diabetes. Autoantibodies are proteins made by your [immune system](#) that attack your healthy tissues and cells by mistake. In rare cases, people with monogenic diabetes may also have autoantibodies in their blood.

Genetic testing

[Genetic testing](#) [NIH](#)  looks for gene variants. Health care professionals use genetic tests to diagnose monogenic diabetes.

Health care professionals may request a genetic test for²

- babies ages 6 months or younger who develop diabetes
- babies who develop diabetes between ages 6 months and 1 year and have some other medical problems
- people who have diabetes at a young age, do not have [overweight](#) or [obesity](#), and have a strong family history of diabetes



If you think you or a member of your family may have a monogenic form of diabetes, talk with your health care professional.

What do I need to know about genetic counseling?

If you, your partner, or family members have monogenic diabetes, you may want to consider [genetic counseling](#) . Genetic counseling can help you decide if you should get tested for monogenic diabetes. Genetic counseling can also help you understand the risk of your children developing this disease. Talk with your health care professional for more information about how genetic counseling could help you, what it may cost, and whether it's covered by your insurance.

How can I manage monogenic diabetes?

How you manage monogenic diabetes will depend on the specific gene variant you have. Your health care professional and diabetes care team will work with you to develop a plan to [manage your diabetes](#). Your health care professional may refer you to, or work with, a medical center that focuses on monogenic diabetes to help manage the disease.

Medicines

Some people with monogenic diabetes may need [diabetes medicines](#). These medicines may include

- [sulfonylureas](#), oral diabetes medicines that help your body make more insulin. These medicines can treat certain forms of MODY and NDM.
- insulin. Some people with monogenic diabetes need to take insulin all their lives. Others may need to take insulin early in life and may be able to switch to sulfonylureas later in life.
- other medicines. Some medicines, called sodium-glucose cotransporter-2 (SGLT2) inhibitors, work in your kidneys to get rid of glucose. Other medicines, called glucagon-like peptide-1 receptor (GLP-1) agonists, increase insulin production and decrease your appetite.

Diabetes medicines and office visits can be expensive. Talk with your health care professional or health care team if you are having trouble affording diabetes care. A health care professional, diabetes educator, or social worker may be able to tell you about resources that provide [financial help for diabetes care](#).

Can I prevent monogenic diabetes?

At this time, monogenic diabetes can't be prevented. However, researchers are working to find possible risk factors and ways to prevent the disease.

Clinical Trials for Monogenic Diabetes

NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for monogenic diabetes?

Clinical trials—and other types of [clinical studies](#) [NIH](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying different aspects of monogenic diabetes, such as ways to improve diagnosis.

[Find out if clinical studies are right for you](#) [NIH](#).

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

Why Should I Join a Clinical Trial?

National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)

From a US national research
authority



Watch on

What clinical studies for monogenic diabetes are looking for participants?

You can find clinical studies on monogenic diabetes at [ClinicalTrials.gov](https://clinicaltrials.gov) [NIH](https://www.nih.gov). In addition to searching for federally funded studies, you can expand or narrow your search to include clinical studies from industry, universities, and individuals; however, the National Institutes of Health does not

review these studies and cannot ensure they are safe for you. Always talk with your primary health care professional before you participate in a clinical study.

What have we learned about monogenic diabetes from NIDDK-funded research?

NIDDK has supported many research projects to learn more about monogenic diabetes

- Progress in Diabetes Genetics in Youth (ProDiGY). This study evaluated how common MODY was among multiethnic youth who had a diagnosis of type 2 diabetes. The study found that 2.8% of the participants had MODY.⁷
- a study with 160 participants showed that people with monogenic diabetes can also have obesity or overweight. They can also have antibodies that are seen in type 1 diabetes.
- Monogenic Diabetes Registry. This database was created in 2008 to collect data related to diagnosis, treatment, and natural history of people with monogenic diabetes.

References

- [1] Salguero MV, Arosemena M, Pollin T, et al. Monogenic forms of diabetes. In: Lawrence JM, Casagrande SS, Herman WH, Wexler DJ, Cefalu WT, eds. *Diabetes in America*. National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK); December 20, 2023.
- [2] Beltrand J, Busiah K, Vaivre-Douret L, et al. Neonatal diabetes mellitus. *Frontiers in Pediatrics*. 2020;8:540718. doi:10.3389/fped.2020.540718
- [3] Blonde L, Umpierrez GE, Reddy SS, et al. American Association of Clinical Endocrinology clinical practice guideline: developing a diabetes mellitus comprehensive care plan—2022 update. *Endocrine Practice*. 2022;28(10):923–1049. doi:10.1016/j.eprac.2022.08.002
- [4] Zhang H, Colclough K, Gloyn AL, Pollin TI. Monogenic diabetes: a gateway to precision medicine in diabetes. *Journal of Clinical Investigation*. 2021;131(3):e142244. doi:10.1172/JCI142244
- [5] ElSayed NA, Aleppo G, Aroda VR, et al. 2. Classification and diagnosis of diabetes: standards of care in diabetes—2023. *Diabetes Care*. 2023;46(suppl 1):S19–S40. doi:10.2337/dc23-S002
- [6] Kant R, Davis A, Verma V. Maturity-onset diabetes of the young: rapid evidence review. *American Family Physician*. 2022;105(2):162–167.
- [7] Todd JN, Kleinberger JW, Zhang H, et al. Monogenic diabetes in youth with presumed type 2 diabetes: results from the Progress in Diabetes Genetics in Youth (ProDiGY) collaboration. *Diabetes Care*. 2021;44(10):2312–2319. doi:10.2337/dc21-0491

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